

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

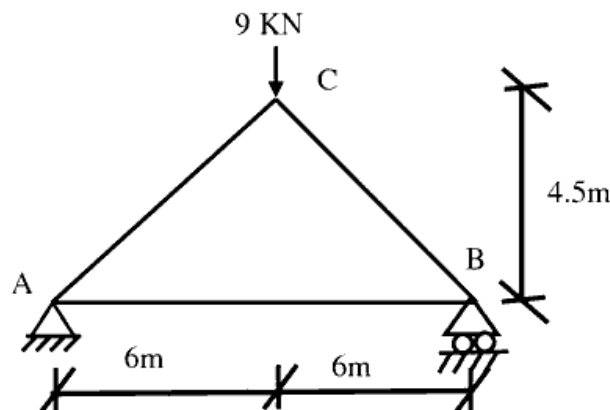
(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

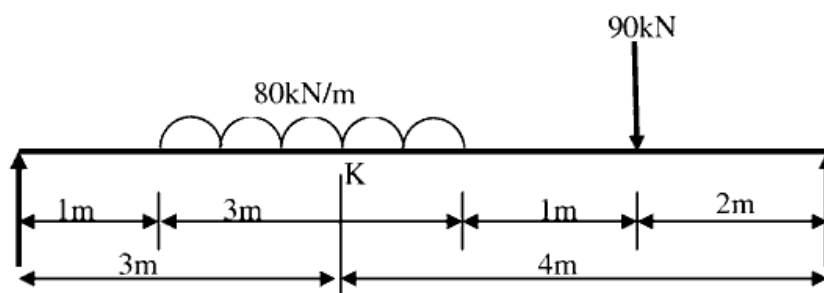
- A) Determine the vertical and horizontal displacement of the joint C of the pin jointed frame as shown in fig. The cross sectional area of AB is 150 mm² and of AC and BC is 200 mm². Assume $E = 200 \text{ kN/mm}^2$.

6



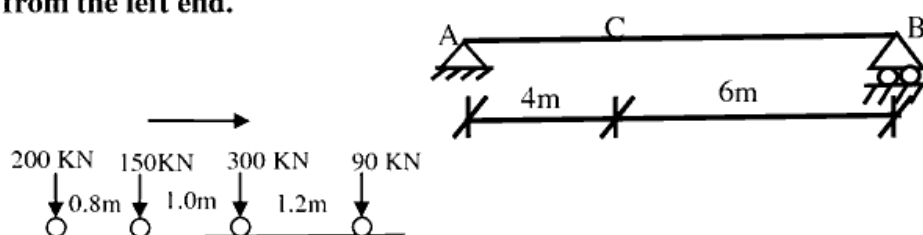
- B) Find the shear force at the section K for the Loaded girder as shown in fig.

6



- C) The Wheel Loads shown in fig. roll over along a beam of span 10 m. Find the maximum bending moment which can occur at a section 4 m from the left end.

6



Q.2 Solve Any Two of the following.

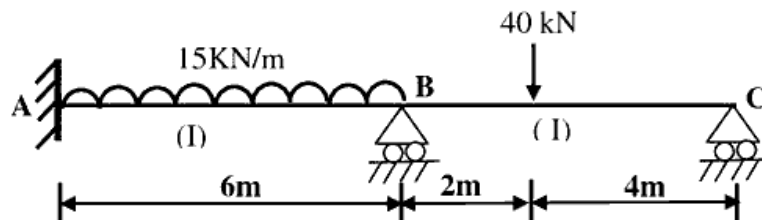
12

- A) A suspension bridge of 100m span has two hinged stiffening girders supported by two cables, having a central dip of 10m. The road way is 6 meters. The dead load on the bridge is 5kN per Sq. meter and the live load is 10kN per Sq. meter, which covers the left of the span. Determine the shear force and bending moment for the girder at 25 m from the left end. Find also the maximum tension in the cable for this position of the live load. 6
- B) A three hinged parabolic arch ABC of span 20m and rise 4m carries uniformly load of 20 KN per meter run on the left half of the span. Find the maximum bending moment for Arch. 6
- C) A two hinged parabolic arch has span of 20m and a rise of 5m and carries a uniformly distributed load of 20kN/m for a distance of 5m from the left end. Determine bending moment, normal thrust and radial shear at a section of the arch 5m from the left end. 6

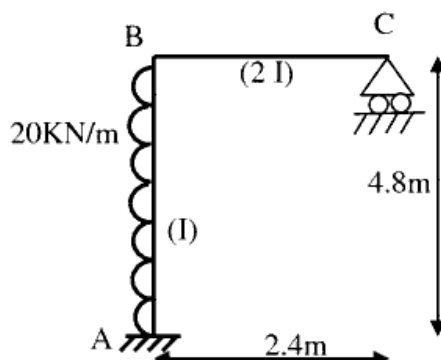
Q.3 Solve Any Two of the following.

12

- A) Analyse the continuous beam as shown in fig. by flexibility approach. If support B sinks by 20mm. Take $EI=3800\text{kN-m}^2$ 6



- B) What is the Relation Between the Stiffness matrix and Flexibility Matrix. 6
- C) Analyse the portal frame as shown in fig. by flexibility approach. 6



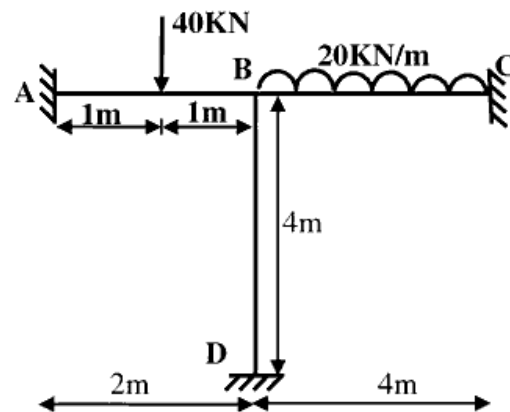
Q.4 Solve Any Two of the following.

12

- A) Differentiate between the force method and displacement method. 6

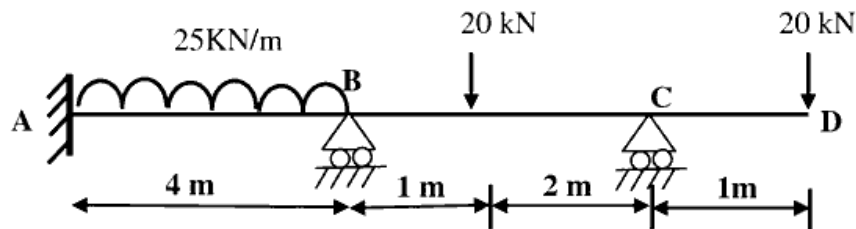
B) Analyse the frame by stiffness method . Draw BMD.EI constnt.

6



C) Analyse the beam as shown in fig. by stiffness matrix method and draw B.M.D.

6



Q. 5 Solve Any Two of the following.

12

A) Write general steps of FEM.

6

B) Explain:

6

- (i) Triangular Element
- (ii) Rectangular element
- (iii) Quadrilateral element

C) Write a note on Pascal's Triangle.

6

*** End ***